CHAPTER 2

PRINCIPLES OF INHERITANCE AND VARIATION

CHAPTER ESSENCE AT A GLANCE:
Mendelian segregation and independent assortment govern transmission genetics, modified by linkage, sex determination, and chromosomal mutations.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Mendelian Laws of Inheritance:** Mendel monohybrid crosses established the Law of Dominance and the Law of Segregation: alleles remain pure and segregate during gamete formation, yielding a 3:1 phenotypic and 1:2:1 genotypic ratio. Dihybrid crosses established the Law of Independent Assortment (9:3:3:1 phenotypic ratio), demonstrating that alleles of distinct gene pairs assort independently into gametes.

• **Non-Mendelian Extensions and Gene Interactions:** Deviations from complete dominance include: Incomplete dominance in Antirrhinum (Snapdragon) and Mirabilis jalapa (red x white yields pink F1, with identical 1:2:1 phenotypic and genotypic F2 ratios); Codominance in human ABO blood groups (IA and IB are codominant, expressing AB phenotype; 6 genotypes produce 4 phenotypes); Pleiotropy where a single gene regulates multiple phenotypic traits (phenylketonuria).

• **Chromosomal Theory and T.H. Morgan Linkage:** Sutton and Boveri formulated the Chromosomal Theory of Inheritance, establishing that chromosomes occur in pairs, segregate during meiosis, and carry Mendelian genes. T.H. Morgan experimentally proved this with Drosophila melanogaster, demonstrating that genes situated on the same chromosome are physically linked, showing low recombination frequencies proportional to distance.

• **Sex Determination and Genetic Disorders:** Sex determination mechanisms: XX-XY in humans and Drosophila; ZZ-ZW in birds (females ZW heterogametic, males ZZ homogametic); Haplodiploidy in honeybees (unfertilized haploid eggs n=16 develop into drones, fertilized diploid eggs 2n=32 into females). Mendelian disorders: Hemophilia (X-linked recessive), Sickle-cell anaemia (autosomal recessive point mutation GAG to GUG at codon 6 of beta-globin chain substituting glutamic acid with valine). Chromosomal disorders: Down's syndrome (trisomy 21), Klinefelter's syndrome (47, XXY), and Turner's syndrome (45, X0).

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Monohybrid Cross Ratios} = \text{Phenotypic: } 3:1 \iff \text{Genotypic: } 1:2:1 \quad [1TT:2Tt:1tt]$$

Mendelian Monohybrid Segregation Ratios

$$\text{Sickle Cell Mutation} = \text{Codon 6: GAG} \rightarrow \text{GUG} \rightarrow \text{Glutamic Acid} \rightarrow \text{Valine in } \beta\text{-chain}$$

Single Base Pair Point Mutation in Hemoglobin Gene

$$\text{Incomplete Dominance} = \text{Phenotype Ratio} = \text{Genotype Ratio} = 1:2:1 \quad \text{(Snapdragon Red:Pink:White)}$$

Equivalence of Phenotypic and Genotypic Ratios in Antirrhinum

$$\text{Aneuploidy Karyotypes} = \text{Down's (Trisomy 21)} + \text{Klinefelter's (47, XXY)} + \text{Turner's (45, X0)}$$

Cytogenetic Karyotypic Deviations in Human Aneuploidy

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In incomplete dominance (Snapdragon), the phenotypic ratio (1 Red : 2 Pink : 1 White) is identical to the genotypic ratio (1 RR : 2 Rr : 1 rr).

• **Alert:** In birds, female heterogamety (ZW) determines sex, while males are homogametic (ZZ), which is opposite to the human XX-XY system.

• **Alert:** In sickle-cell anaemia, a single point mutation at the sixth position of the beta-globin chain substitutes Glutamic acid (hydrophilic) with Valine (hydrophobic).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is a test cross? Show how a test cross is used to determine whether a tall pea plant is homozygous or heterozygous. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Definition: A test cross is a genetic cross between an individual exhibiting a dominant phenotype with a homozygous recessive parent. **[1 Mark]**

• Interpretation: If the unknown tall plant is homozygous (TT), all progeny will be tall; if heterozygous (Tt), tall and dwarf progeny appear in a 1:1 ratio. **[1 Mark]**

2. In Snapdragon (Antirrhinum majus), crossing true-breeding red-flowered plants with white-flowered plants yields all pink-flowered F1 progeny. State the inheritance pattern and provide the phenotypic and genotypic ratios of the F2 generation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Inheritance pattern: Incomplete dominance, where neither allele is completely dominant over the other, producing an intermediate blend. **[1 Mark]**

• F2 ratios: Both the phenotypic ratio and genotypic ratio are 1 Red (RR) : 2 Pink (Rr) : 1 White (rr) (ratio 1:2:1). **[1 Mark]**

3. Explain the molecular basis of Sickle-cell anaemia, detailing the specific nucleotide point mutation and amino acid substitution in the beta-globin chain. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Inheritance: An autosomal recessive disorder transmitted when both parents are carriers (heterozygous HbA HbS). **[1 Mark]**

• Point mutation: A single base substitution from adenine to thymine (GAG to GUG) at the sixth codon of the beta-globin gene on chromosome 11. **[1 Mark]**

• Amino acid substitution: Results in the replacement of polar glutamic acid by non-polar valine at the sixth position, causing mutant hemoglobin to polymerize under low oxygen into rigid sickle shapes. **[1 Mark]**

4. Differentiate between Down's syndrome, Klinefelter's syndrome, and Turner's syndrome with respect to their chromosomal karyotypes and diagnostic clinical symptoms. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Down's syndrome: Karyotype has 47 chromosomes (Trisomy of autosome 21); symptoms include short stature, furrowed tongue, partially open mouth, broad palm with palm crease, and mental retardation. **[1 Mark]**

• Klinefelter's syndrome: Karyotype has 47 chromosomes (44 autosomes + XXY); sterile male with tall stature, overall masculine development, but feminine development like breast enlargement (gynecomastia). **[1 Mark]**

• Turner's syndrome: Karyotype has 45 chromosomes (44 autosomes + X0); sterile female with short stature, webbed neck, rudimentary ovaries, and lack of secondary sexual characteristics. **[1 Mark]**

5. Explain the mechanisms of sex determination in: (a) Humans (XX-XY), (b) Birds (ZZ-ZW), and (c) Honeybees (Haplodiploidy). [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) Humans: Male heterogamety; females possess XX chromosomes producing identical X gametes, while males possess XY producing 50% X and 50% Y sperms; father's sperm determines sex. **[1 Mark]**

• (b) Birds: Female heterogamety; males possess ZZ homogametic chromosomes, while females possess ZW heterogametic chromosomes; mother's ovum determines sex. **[1 Mark]**

• (c) Honeybees: Haplodiploid system; unfertilized haploid eggs (n=16) develop into fertile males (drones) by parthenogenesis, while fertilized diploid eggs (2n=32) develop into females (queens or worker bees). **[1 Mark]**

6. Why did T.H. Morgan choose the fruit fly (Drosophila melanogaster) for his experimental research on linkage and crossing over? State any four advantages. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Could be grown easily on simple synthetic medium in the laboratory. **[0.75 Mark]**

• Complete life cycle is completed in about two weeks, yielding rapid experimental turnover. **[0.75 Mark]**

• A single mating produces a large number of progeny flies for statistical analysis. **[0.75 Mark]**

• Clear sexual dimorphism (males easily distinguished from larger females) and numerous distinct hereditary variations visible under low-power microscopes. **[0.75 Mark]**

7. A hemophilic man marries a phenotypically normal woman whose father was hemophilic. Determine the expected probabilities of their sons and daughters being hemophilic or carriers, providing a genetic cross. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Parental genotypes: Hemophilic father = XhY; Mother is phenotypically normal but must have received Xh from her hemophilic father, so her genotype is heterozygous carrier XH Xh. **[1 Mark]**

• Gametes: Father produces Xh and Y sperms; Mother produces XH and Xh ova. **[1 Mark]**

• Offspring genotypes: (1) XH Xh (carrier daughter), (2) Xh Xh (hemophilic daughter), (3) XH Y (normal son), (4) Xh Y (hemophilic son). **[1 Mark]**

• Probabilities: Among daughters: 50% carrier, 50% hemophilic. Among sons: 50% normal, 50% hemophilic. Overall offspring: 50% hemophilic, 25% carrier, 25% normal. **[1 Mark]**